AI FOR GOOD · HACKATHON 6 · SDG 9

Heat Pump Pulse

A live read of how the UK talks about heat pumps, so an advice team knows which worry to answer first.

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THE USER AND THEIR DECISION

Built for the Energy Saving Trust's advice team

Who

The UK team that writes heat pump advice for households and already fights myths about them.

Their decision, every week

1. Which worry to answer first: cost, installers, cold, noise or myths.

2. When a news story calls for extra explanation.

Why it matters: 24% think heat pumps cost more to run than a gas boiler (Opinium, 2023) · 56% find them "too expensive for most people" (Savanta, 2025)

HOW IT WORKS

Three scripts, running non-stop

Bluesky stream
Every new English post about heat pumps, live

→

Collector
Skips bots, stores text and time, no names

→

Transformer
twitter-roberta scores the tone of each post

→

Dashboard
Tone over time, themes, and the posts behind it

668
posts collected

3.5 days
without gaps

620
scored by the model

Snapshot 15 Oct 2026, 20:47 UK time · 47 posts excluded as automated, duplicate or too short

LIVE DEMO

The tone, every six hours



Backup screenshot · the dashboard refreshes itself every 2 minutes

BEHIND THE BIGGEST DIP

The morning after the grant-cut news

Tue 13 Oct, 06:00–12:00 UK

−0.29

average tone, 58 posts: the lowest period in our data

“Together with more gas & oil … are looking to cut back on heat pump subsidy to fund defence … feckless!”

Model: negative, −0.84

“… Labour is preparing to slash hundreds of millions of pounds from the Boiler Upgrade Scheme to help fund a £15bn boost to UK defence spending.”

Model: negative, −0.55

12 Oct 2026: the Guardian and Politico report that ministers are considering cutting the Boiler Upgrade Scheme (up to £7,500 per heat pump)

THE HONEST TEST

Our labels against two models

Transformer (twitter-roberta)

72%

agrees with us on 36 of 50

VADER (word-list baseline)

48%

calls 10 of 18 negative posts positive

Transformer: where it goes wrong

Ours ↓ · model →	negative	neutral	positive
negative (18)	13 ✓	4	1
neutral (22)	4	16 ✓	2
positive (10)	0	3	7 ✓

✓ = agreement (36 of 50). Most mistakes are one step off; only **1** negative post was called positive.

We hand the user the transformer. VADER would show a calm line exactly when people are worried.

50 random posts labelled by Majed before seeing any model output · full table in test_table.csv

THE FAILURE THAT TAUGHT US THE MOST

41%

of the posts the model scored negative were written by people in favour of heat pumps. Only 2 of 17 were against.

At first we read the dip as resistance. It was anger about a cut. **Tone is not opinion**, so the dashboard now says so at the top, and every dip opens to its posts.

ETHICAL REFLECTION

Real people, a partial picture

Who is missing

Only English-speaking Bluesky users, a fairly left-leaning group. Older people and households who never post are not in the line.

What the model misreads

Tone, not opinion: 41% of “negative” posts came from supporters. It also misses sarcasm and reads only English.

Who could be harmed

Authors can be found by searching a sentence. A team that trusts the line too much answers the wrong worry.

What we did about it

No names stored · deleted posts removed · bots and spam excluded · post counts and a warning on the dashboard · private repository

USE IT WITH CARE

What the pulse can and cannot tell you

It can

Show when people get upset, and which worry grows, early enough to answer it.

It cannot

Say who is for or against heat pumps, or speak for households who never post.

Thank you. Questions?